

Smooth six-functor formalisms

Workshop “Alpbach 2026”

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ETH Zürich

One reason why you should care

Counting rational points

Let X be a d -dimensional variety over a finite field $k = \mathbb{F}_q$. Given that you are here, you might care about $|X(k)|$. These numbers are related to deep results in arithmetic geometry. Notably, the Weil conjectures.

Weil famously suggested that his conjectures might be related to some imaginary cohomology theory, that later became known as ℓ -adic cohomology:

$$|X(k)| = \sum_{i=0}^{2d} (-1)^i \operatorname{tr} \left(\operatorname{Frob}_k \mid H_c^i(X_{\bar{k}}, \mathbb{Q}_\ell) \right).$$

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Climbing the categorical ladder

Using these cohomology groups, Grothendieck proved a large part of Weil's conjectures. Then, Deligne proved a particularly elegant generalization of the remaining one.

He defined a notion of weights for a complex in $D_c^b(X_{\bar{k}}, \mathbb{Q}_\ell)$ and proved that the functor $Rf_!$ preserves complexes of weights $\leq w$.

Not only this generalizes the remaining conjecture, but it also gives the following useful bound:

$$||X(\mathbb{F}_q)| - q^{nd}| \leq Cq^{n(2d-1)/2}.$$

$$\begin{array}{c} D_c^b(X_{\bar{k}}, \mathbb{Q}_\ell) \\ \downarrow \text{Hom} \\ H_c^i(X_{\bar{k}}, \mathbb{Q}_\ell) \\ \downarrow \text{tr}(\text{Frob}) \\ |X(k)| \end{array}$$

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Correspondences and six-functor formalisms

The *yoga* of six operations

For a long time, six-functor formalisms had no rigorous definition; **you knew one when you saw it**. Here's what happens in all examples:

- You have a certain (possibly ∞ -)category of spaces $\mathcal{C}$.
- For every space X in $\mathcal{C}$, there is a closed symmetric monoidal stable ∞ -category $D(X)$ (i.e., we have $\otimes$ and Hom functors, which are adjoint).
- For every morphism $f: X \rightarrow Y$ in $\mathcal{C}$, we have a symmetric monoidal functor $f^*: D(Y) \rightarrow D(X)$ that admits a right adjoint f_* .
- There exists a collection $\mathcal{I}$ of morphisms in $\mathcal{C}$ such that, for every morphism $f: X \rightarrow Y$ in $\mathcal{I}$, we have adjoint functors

$$f^*: D(Y) \rightleftarrows D(X)$$

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- There exists a collection E of morphisms in C such that, for every morphism $f: X \rightarrow S$ in E , we have adjoint functors

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Base change and the projection formula

These functors should satisfy certain compatibilities. Notably:

- (Projection formula) For f in E , there exists a natural isomorphism

$$f_!(- \otimes -) \simeq f_!(- \otimes f^* -).$$

- (Base change) Given a cartesian diagram in $\mathcal{C}$

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{g} & S, \end{array}$$

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Proper maps and open immersions

We also expect classes of morphisms $J, P \subset E$ such that, for $f \in E$:

- If Δ_f lies in P , there exists a natural transformation $f_! \rightarrow f_*$. It is an isomorphism if $f \in P$.
- If Δ_f lies in J , there exists a natural transformation $f^* \rightarrow f^!$. It is an isomorphism if $f \in J$.

In motivic settings one often adds even more axioms, but those are the axioms satisfied by every six-functor formalism that I know.

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How to define a six-functor formalism? Property vs data

Suppose we want to build a six-functor formalism from scratch: what **data** do we need to provide?

- The co-categories $D(X)$, with their symmetric monoidal structures. (Being stable and admitting a inner Hom are properties.)
- Functors $f^*: D(S) \rightarrow D(X)$, as well as their symmetric monoidal structures. (Admitting a right adjoint f_* is a property.)
- Functors $f_*: D(X) \rightarrow D(S)$ (admitting a right adjoint f^* is a property.)
- The projection formula: $\alpha \otimes f_* \beta \xrightarrow{\sim} f_* f^* \alpha \otimes \beta$
- The base change compatibility: $g^* f_* \xrightarrow{\sim} f'_* g'^*$
- The formal adjunctions, which can be provided as natural transformations $f^* \dashv f_*$ and $f_* \dashv f^*$.

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- Functors $f_! : D(X) \rightarrow D(S)$. (Admitting a right adjoint $f^!$ is a property.)
- The projection formula $f_! \otimes - \rightarrow f_!(- \otimes f^* -)$.
- The base change isomorphisms $g^* f_! \rightarrow f'_! g'^*$.
- The forget-supports maps, which can be encoded as natural transformations $f^* f_! \rightarrow \text{id}$ and $f_! f^* \rightarrow \text{id}$.

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This is not enough!

- If $h = g \circ f$, we should provide natural isomorphisms $h^* \rightarrow f^*g^*$ and $h_! \rightarrow g_!f_!$.
- These isomorphisms should be compatible with the projection formula:

$$\begin{array}{ccccc} h_!(- \otimes -) & \longrightarrow & h_!(- \otimes h^*-) & & \\ \downarrow & & \downarrow & \searrow & \\ g_!f_!(- \otimes -) & & g_!f_!(- \otimes h^*-) & & h_!(- \otimes f^*g^*-). \\ \downarrow & & \downarrow & \swarrow & \\ g_!(f_!(- \otimes g^*-)) & \longrightarrow & g_!f_!(- \otimes f^*g^*-) & & \end{array}$$

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Not even this is enough!

- The composition isomorphisms should be compatible with the base change isomorphisms. (I'll spare you the diagram...)
- The base change isomorphisms should be compatible with the projection formula.
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Note that, since we are working with ∞ -categories, commuting is not a **property** that some diagram may satisfy; it is extra **data** that has to be provided.

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Lurie's brilliant idea: correspondences

Suppose that $\mathcal{C}$ is a 1-category admitting fiber products and that $J, P \subset E$ are stable under composition and base change. We will construct a 2-category $\text{Corr}_{J,P}(\mathcal{C}, E)$ of **correspondences**.

The objects of $\text{Corr}_{J,P}(\mathcal{C}, E)$ are the same as the objects of $\mathcal{C}$. A morphism $X \rightarrow Y$ in $\text{Corr}_{J,P}(\mathcal{C}, E)$ amounts to a diagram



where $W \rightarrow Y$ lies in E .

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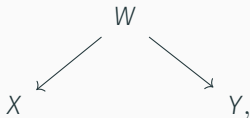


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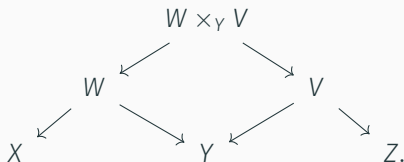
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Correspondences II

To compose $X \leftarrow W \rightarrow Y$ and $Y \leftarrow V \rightarrow Z$, we take a fiber product:

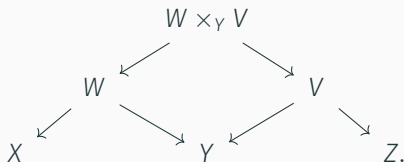


Finally, a 2-morphism $(X \leftarrow W_1 \rightarrow Y) \rightarrow (X \leftarrow W_2 \rightarrow Y)$ is a diagram

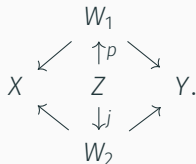


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Three-functor formalisms

The 2-category $\text{Corr}_{J,P}(C, E)$ has a symmetric monoidal structure, where $X \otimes Y := X \times Y$.

Definition (Lurie, Liu-Zheng)

A **three-functor formalism** is a lax symmetric monoidal functor

$$D: \text{Corr}_{J,P}(C, E) \rightarrow \text{Cat}_{\text{st}}.$$

Here, the adjective **lax** means that instead of having isomorphisms $D(X) \times D(Y) \simeq D(X \times Y)$, we only have a natural transformation

$$\boxtimes: D(-) \times D(-) \rightarrow D(- \times -).$$

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Let's unpack this definition

Given a three-functor formalism $D: \text{Corr}_{J,P}(C, E) \rightarrow \text{Cat}_{\text{st}}$, we define:

- $f^*: D(Y) \rightarrow D(X)$ as the image of $Y \xleftarrow{f} X = X$ by D ;
- $f_!: D(X) \rightarrow D(Y)$ as the image of $X = X \xrightarrow{f} Y$ by D ;
- $\otimes$ as the composition

$$D(X) \times D(X) \xrightarrow{\boxtimes} D(X \times X) \xrightarrow{\Delta_X^*} D(X),$$

where $\Delta_X: X \rightarrow X \times X$ is the diagonal map.

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Definition (Lurie, Liu-Zheng)

A **six-functor formalism** is a three-functor formalism in which the functors f^* , $f_!$ and $\otimes$ admit right adjoints.

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Given a three-functor formalism $D: \text{Corr}_{J,P}(C, E) \rightarrow \text{Cat}_{\text{st}}$, we define:

- $f^*: D(Y) \rightarrow D(X)$ as the image of $Y \xleftarrow{f} X = X$ by D ;
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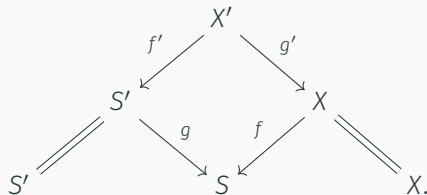


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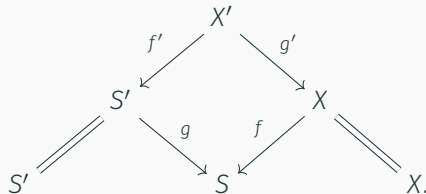


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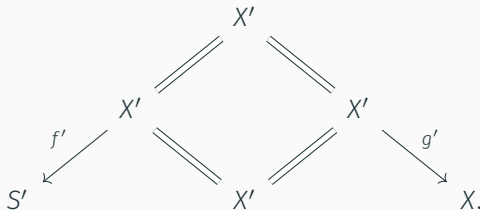
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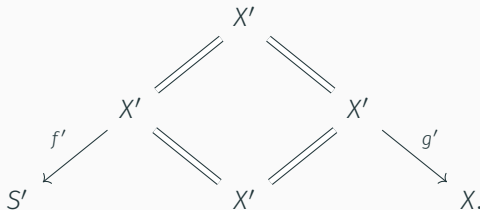


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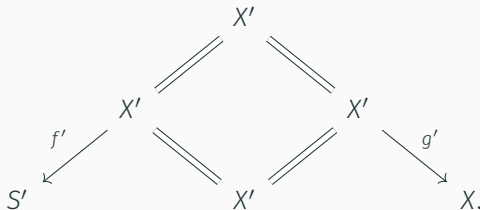


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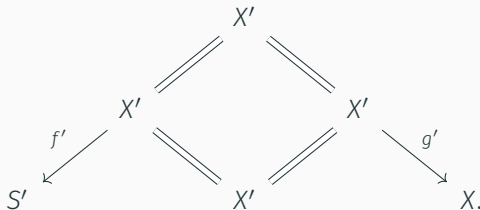


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Deligne's idea to the rescue

Since a functor $D: \text{Corr}_{J,P}(C, E) \rightarrow \text{Cat}_{\text{st}}$ encodes all the data on a six-functor formalism... **constructing it must be impossible, right?**

It's often simple to construct a functor $C^{\text{op}} \rightarrow \text{CAlg}(\text{Cat}_{\text{st}})$ encoding the functors f^* and $\otimes$. If every map $f \in E$ factors as $f = p \circ j$, for some $j \in J$ and $p \in P$, we may try to construct $p_!$ and $j_!$ separately.

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In general, base change and the projection formula are relations between the functors $f_!$ and f^* , which need not be adjoint in any direction. When they are, the adjunction provides natural maps that can be checked to be isomorphisms.

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Six-functor formalisms on smooth schemes

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Let X be a smooth algebraic variety over $\mathbb{C}$. A theorem of Grothendieck asserts that

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Both cohomology theories have natural coefficients: holonomic $\mathcal{D}$ -modules and constructible sheaves. Grothendieck's theorem can then be enhanced into **an isomorphism of six-functor formalisms**

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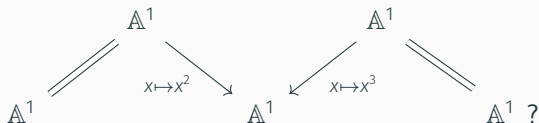
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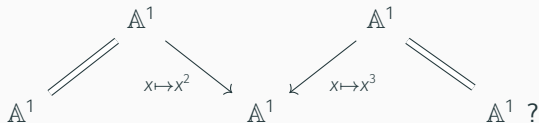
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Fix a base scheme S . A **three-functor seed on Sm/S** is a functor $D: (\mathrm{Sm}/S)^{\mathrm{op}} \rightarrow \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ such that:

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The locality property suggests a way of extending local three-functor seeds to singular schemes! For a (possibly singular) quasi-projective scheme X , there exists a closed immersion

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where $\tilde{X}$ is smooth (an open subscheme of $\mathbb{P}^n$).

Then, denoting by $j: U \hookrightarrow \tilde{X}$ its complement, we may **define**

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Given a general (qcqs) S -scheme X , one may:

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to define $D(X)$.

Theorem (Gallauer–R.)

This construction extends to a three-functor formalism on Sch/S , and gives an equivalence of $(\infty, 2)$ -categories

$$\mathrm{TFS}_{\mathrm{inv}, \mathrm{loc}}(\mathrm{Sm}/S; J, P) \simeq \mathrm{TFF}_{\mathrm{cont}, \mathrm{loc}}(\mathrm{Sch}/S; E, J, P),$$

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Ideas of the proof: the category Em/S

Let QProj/S be the category of quasi-projective S -schemes, and let Em/S be the following category:

- its objects are closed immersions $X \hookrightarrow \tilde{X}$ in QProj/S with $\tilde{X}$ smooth;
- a morphism $(X \hookrightarrow \tilde{X}) \rightarrow (Y \hookrightarrow \tilde{Y})$ is a commutative square

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The category Em/S admits natural functors:

$$\begin{array}{ccccc} \text{Sm}/S & \xrightarrow{d} & \text{Em}/S & \xrightarrow{s} & \text{QProj}/S \\ X & \longmapsto & (X \xrightarrow{\text{id}} X) & & \\ & & (X \hookrightarrow \tilde{X}) & \longmapsto & X. \end{array}$$

Ideas of the proof: the category Em/S

Let QProj/S be the category of quasi-projective S -schemes, and let Em/S be the following category:

- its objects are closed immersions $X \hookrightarrow \tilde{X}$ in QProj/S with $\tilde{X}$ smooth;
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Ideas of the proof: the big picture

Let D be an ambient-invariant local three-functor seed on Sm/S . To obtain a three-functor seed on Sch/S , we extend D progressively.

$$\begin{array}{ccccccc} (\mathrm{Sm}/S)^{\mathrm{op}} & \xrightarrow{d} & (\mathrm{Em}/S)^{\mathrm{op}} & \xrightarrow{s} & (\mathrm{QProj}/S)^{\mathrm{op}} & \hookrightarrow & (\mathrm{FP}/S)^{\mathrm{op}} \hookrightarrow (\mathrm{Sch}/S)^{\mathrm{op}} \\ \downarrow D & & \swarrow (1) & \swarrow (2) & \swarrow (3) & \swarrow (4) & \\ \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}}) & & \swarrow & \swarrow & \swarrow & \swarrow & \end{array}$$

Step (1) is a non-trivial geometric statement, while step (2) is a non-trivial categorical statement. Steps (3) and (4) are simple dévissages, which I'll ignore.

Having finished these four extensions, the result follows from Cnossen–Lenz–Linskens' theorem.

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Ideas of the proof: the geometric step

Extending $D: (\text{Sm}/S)^{\text{op}} \rightarrow \text{CAlg}(\text{Cat}_{\text{st}})$ to Em/S is easy: for an object $X \hookrightarrow \tilde{X}$ of Em/S with complement $j: U \hookrightarrow \tilde{X}$, we define

$$\bar{D}(X \hookrightarrow \tilde{X}) := D(\tilde{X})/j_! D(U).$$

With standard tools, one checks that this is indeed a functor $\bar{D}: (\text{Em}/S)^{\text{op}} \rightarrow \text{CAlg}(\text{Cat}_{\text{st}})$.

The important part of this step is verifying that $\bar{D}$ satisfies the axioms of a three-functor seed “on the source”.

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Ideas of the proof: the geometric step II

For example, let (f, g) be the morphism in Em/S represented by the diagram

$$\begin{array}{ccc} X & \hookrightarrow & \tilde{X} \\ f \downarrow & & \downarrow g \\ Y & \hookrightarrow & \tilde{Y}. \end{array}$$

If f is an open immersion, we have to show that

$$(f, g)^*: \overline{D}(Y \hookrightarrow \tilde{Y}) \rightarrow \overline{D}(X \hookrightarrow \tilde{X})$$

is naturally isomorphic to $(f, \bar{g})^*$, where $\bar{g}$ is also an open immersion.

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Categorical digression

Recall that a functor $F: C \rightarrow D$ is said to be a **localization** if

$$\mathrm{Fun}(D, E) \xrightarrow{- \circ F} \mathrm{Fun}_W(C, E)$$

is an equivalence for all categories E , where

$$W := \{f \in \mathrm{Mor}(C) \mid F(f) \text{ is an isomorphism}\}$$

and $\mathrm{Fun}_W(C, E)$ is the category of functors $G: C \rightarrow E$ such that $G(f)$ is an isomorphism for all $f \in W$.

If F satisfies this universal property with respect to all ∞ -categories E , we say that F is an **∞ -categorical localization**.

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Proposition

The functor $s: \mathrm{Em}/S \rightarrow \mathrm{QProj}/S$ is an ∞ -categorical localization.

The proof is based on “Hinich’s Key Lemma”: if $F: C \rightarrow D$ is such that $\mathrm{Fun}([n], F)$ has weakly contractible fibers for all $n \geq 0$, then F is an ∞ -categorical localization.

As a result, our functor $\bar{D}: (\mathrm{Em}/S)^{\mathrm{op}} \rightarrow \mathrm{CAlg}(\mathrm{Cat}_{\mathrm{st}})$ extends uniquely to QProj/S .

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Questions?